Adrian Marin Mag

Mathematical Geophysicist | Scientific Computing & Data Science

Oxford, United Kingdom
adrian-mag.github.io
ORCID: 0009-0001-1424-6664

Professional Summary

Oxford PhD candidate in mathematical geophysics who develops statistical and physics-based models to turn incomplete and noisy geophysical data into reliable estimates with quantified uncertainty. Experienced in uncertainty quantification, predictive modelling, optimisation, and tested Python research software. Published author and university tutor accustomed to owning independent technical work, collaborating across disciplines, and explaining complex results to technical and non-technical audiences.

Relevant Capabilities

Inference for decisions Turn sparse and noisy geophysical observations into targeted estimates of subsurface properties, quantify their uncertainty, and distinguish conclusions supported by data from those driven by modelling assumptions.

Model development Build and critically evaluate statistical, Bayesian, and physics-based models; test sensitivity to assumptions and data limitations, interpret predictions, and communicate their practical limitations clearly.

Scientific software Develop tested, documented Python tools and reproducible workflows for spatial and three-dimensional geophysical analysis, using Git and continuous integration to make methods reliable, reviewable, and reusable.

Technical delivery Own long-running research and software projects from problem definition through implementation, publication, and release; collaborate across institutions and explain technical trade-offs to audiences with varied backgrounds.

Experience

2022–Present **PhD Researcher, Mathematical Geophysics**, *University of Oxford*, Oxford.

- Develop methods that turn sparse and noisy seismic observations into targeted estimates of subsurface properties, with uncertainty reported alongside each result.
- Unified two previously separate approaches to linear inference in a peer-reviewed framework and released accompanying tested Python software, enabling local averages, gradients, and spectral properties to be estimated within one reproducible workflow.
- Develop mathematical and computational methods for testing how physical constraints and prior assumptions change an inference, making clear which conclusions are supported by data and which depend on modelling choices.
- Built Python workflows that convert AxiSEM3D wavefield simulations into three-dimensional finite-frequency sensitivity kernels; the workflows are included in the official AxiSEM3D example gallery.
- Collaborate with researchers in Oxford, Cambridge, and Strasbourg, delivering peer-reviewed research, open-source software, documentation, presentations, and worked computational examples.

2024–2026 **Python Course Demonstrator**, *University of Oxford*, Oxford.
Supported intermediate and advanced scientific-Python courses for environmental scientists, covering NumPy, SciPy, Matplotlib, VS Code, Linux, Numba, Dask, Xarray, and Cartopy; guided practical exercises, debugging, and reproducible workflow development.

2023–2026 **College Tutor, Mathematics**, *Exeter & Worcester Colleges, University of Oxford*, Oxford.
Planned and delivered tutorials in calculus, statistics, vector calculus, and continuum mechanics; translated technical material into clear explanations and provided individual feedback.

2020–2021 **Tutor and Teaching Developer**, *Notebook Tutors / University College London*, London.
Taught mathematics and physics, created interactive MATLAB notebooks for numerical methods, and developed a randomised online assessment bank for global geophysics.

Selected Open-Source Software

AxiSEM3D kernels Built reproducible adjoint workflows that convert AxiSEM3D simulations into three-dimensional finite-frequency sensitivity kernels for full-waveform inversion; contributions are included in the official AxiSEM3D example gallery.

SOLA–DLI Developed and released reproducible Python software for a peer-reviewed inference framework, with numerical examples, documentation, automated testing, and continuous integration.

pygeoinf Developed the core convex-geometry framework for encoding prior information about subsurface models as convex sets, including dual support equations, reusable support-function and convex-set classes, optimisation algorithms, and one-, two-, and three-dimensional visualisation. This makes prior dependence in linear-inference results explicit.

Technical leadership Led development of a Python tool for ray-theoretical seismic sensitivity kernels, contributing most of the core code and supervising a master's student. Set the technical direction, reviewed proposed features, and advised their implementation.

Technical Skills

Programming	Python; working knowledge of MATLAB and C++
Data and scientific Python	NumPy, SciPy, pandas, Xarray, Matplotlib, Cartopy, Jupyter, pytest, Sphinx
Engineering	Git, GitHub Actions, packaging, unit testing, documentation, CI workflows, Linux
Computing environments	High-performance computing on ARC and ARCHER
Modelling	Applied statistics, Bayesian inference, Gaussian processes, uncertainty quantification, inverse problems, regularisation, convex optimisation, adjoint methods
Spatial and physical data	Seismic tomography, 2D/3D sensitivity kernels, spherical-harmonic analysis, wave propagation, travel-time modelling, scientific visualisation
Responsible AI	Auditable AI-assisted research and software workflows, with source verification, sandboxed execution, human scientific review, and version-controlled provenance

Education

2022–Present **PhD in Mathematical Geophysics**, *University of Oxford*, Oxford.
Expected completion: December 2026. Thesis: *Building Inference from the Ground Up: A Framework for Linear Inverse Problems and Property Inference*. Develops practical mathematical and computational tools for determining what incomplete geophysical data can reliably reveal, how assumptions affect predictions, and how uncertainty should be interpreted.

2018–2022 **MSci in Geophysics, First-Class Honours (82%)**, *University College London*, London.
Thesis: *Towards building 3D global seismic tomography models using proximal methods*. Developed numerical methods for reconstructing three-dimensional Earth structure from seismic observations.

Publications and Manuscripts

Published — Mag, A. M., Zaroli, C., & Koelemeijer, P. (2025). Bridging the gap between SOLA and deterministic linear inferences in the context of seismic tomography. *Geophysical Journal International*, 242(1), ggaf131.

In preparation — Mag, A. M., Al-Attar, D., Isaac, J., & Koelemeijer, P. *A geometric framework for multi-parameter linear inference*. Manuscript in preparation.

In preparation — Mag, A. M., Zaroli, C., Koelemeijer, P., & Scivier, S. *Property-based probabilistic linear inference in function space for geophysical inverse problems*. Manuscript in preparation.

Selected Technical Writing

Think First, Discretize Later — Function-space formulation, model geometry, adjoints, and discretisation in linear inverse problems.

My Take on SOLA — Sensitivity and resolving kernels, inference targets, noise, and the distinction between propagated noise and posterior uncertainty.

Bayes, Measure-Theoretically — Posterior kernels and measure-theoretic Bayesian inversion developed from a finite example.

The Road to Conjugate Gradients — A function-analytic route from gradient descent to the conjugate-gradient method.

The Machine Around the Model — Auditable agentic research workflows, tool use, context, sandboxing, and verification.

Conferences and Presentations

- 2026 Poster: *Think First, Discretize Later*. European Geosciences Union General Assembly, Vienna.
- 2025 Talk: *Beyond inversions: the flexibility of linear inference methods*. PGRiP, Oxford.
- 2025 Poster: *Combining SOLA and Deterministic Linear Inferences*. Inge Lehmann Symposium, Copenhagen.
- 2025 Poster: *SOLA inferences and normal modes*. MODES/Deep Earth Meeting, Oxford–Cambridge.
- 2024 Poster: *Constraining Earth model properties through Backus–Gilbert SOLA inferences*. UKSEDI, Leeds; BSM version and supplementary material, Reading.
- 2023 Poster: *Mapping the Topography of Earth's Core*. PGRiP, Edinburgh.

Additional Information

Work authorisation — United Kingdom: settled status; authorised to work without employer sponsorship.

Travel — Romanian passport holder; willing to travel internationally.